

System Design Master (Senior Engineer / Tech Lead)

How to Use This Document

- Use this file as the single source of truth.
- For each question, fill all template sections.
- Diagrams are referenced as images under diagrams/.

Export (DOCX/PDF)

See the **Export Commands** section at the end of this file.

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3. Design an API Gateway + AuthN/AuthZ (JWT/OAuth2)
4. Design a Feature Flag / Config Service
5. Design a Logging Platform (collection → storage → search)
6. Design a Metrics/Monitoring Platform (ingestion + aggregation)

B. Kafka / Event-Driven Systems

7. Design a Notification System (email/SMS/push; retries/DLQ)
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10. Design Outbox + CDC Integration Pattern
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Standard Template (Use for Every Question)

1) Problem Statement

2) Features / Functional Requirements

3) Non-Functional Requirements

- Availability:
- Latency (p95/p99):
- Scale (DAU/MAU, peak QPS):
- Consistency:
- Durability:
- Cost constraints:

4) Assumptions + Back-of-the-envelope Sizing

- Peak QPS:
- Storage/day:
- Bandwidth:
- Hot keys / hotspots:

5) High-Level Architecture

6) APIs

7) Data Model

8) Kafka / Async Design (if applicable)

- Topics (names, partitions, retention):
- Ordering guarantees:
- Consumer groups:
- Retry strategy:
- DLQ:
- Idempotency/dedup:
- Schema evolution:

9) Diagrams

- Architecture diagram:
- Key flow diagram (sequence):

10) Key Flows

- Write path:
- Read path:
- Async path:

11) Advantages

12) Disadvantages / Risks

13) Tradeoffs + Alternatives

14) Failure Modes + Mitigations

15) Observability (SLIs/SLOs)

- Metrics:
- Logs:
- Traces:
- Alerts:

16) Security + Compliance

- AuthN/AuthZ:
- PII handling:
- Encryption:

- Auditing:

17) Kubernetes Deployment View

- Workloads:
- HPA:
- Requests/limits:
- Probes:
- Rollout strategy:
- PDB:

18) Capacity Planning + Cost

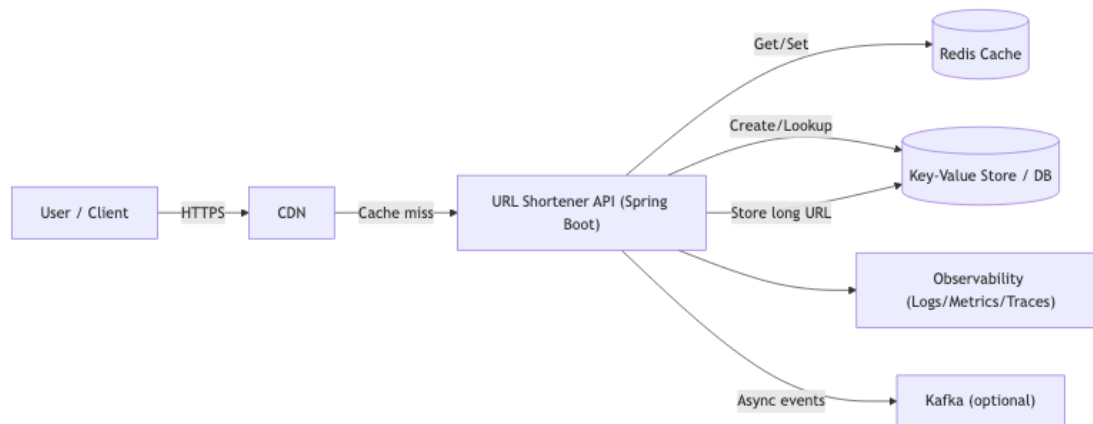
19) Evolution Plan (v1 → v2 → v3)

Questions

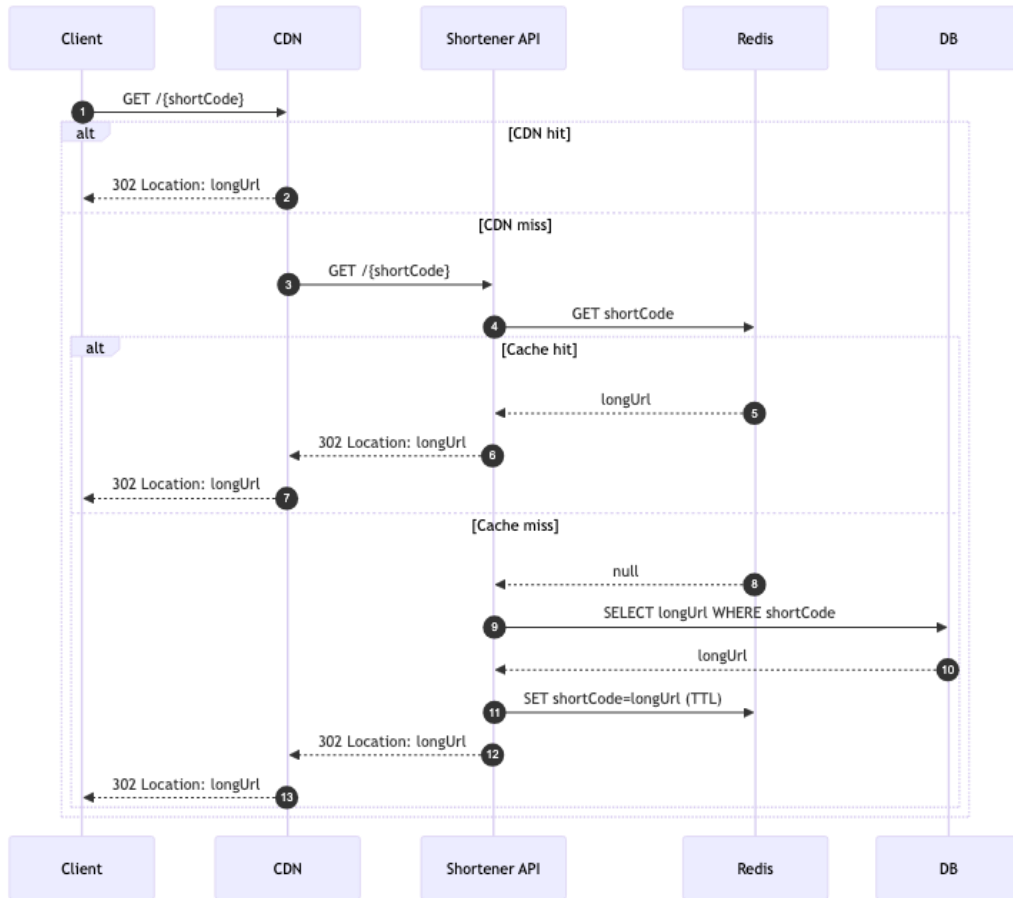
Q01: Design a URL Shortener

[Fill using the Standard Template]

Diagrams



Q01 Architecture

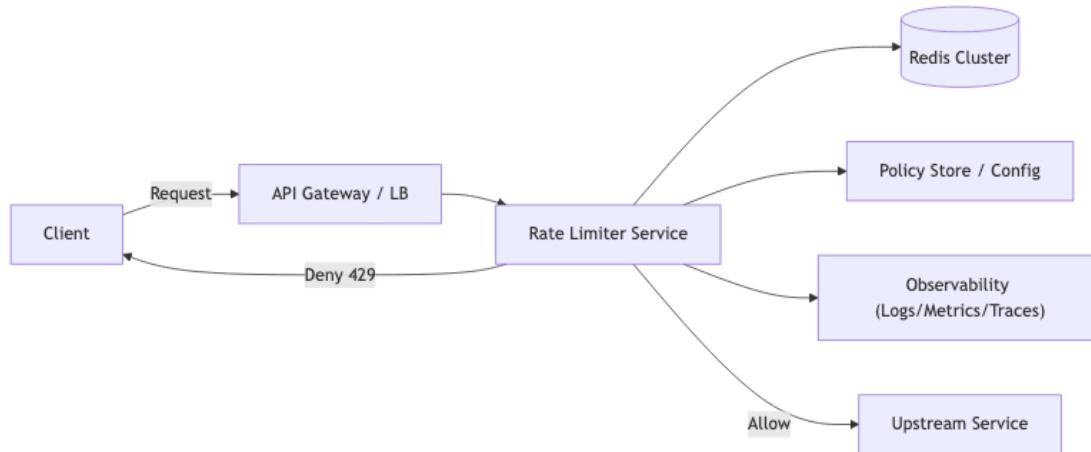


Q01 Key Flow

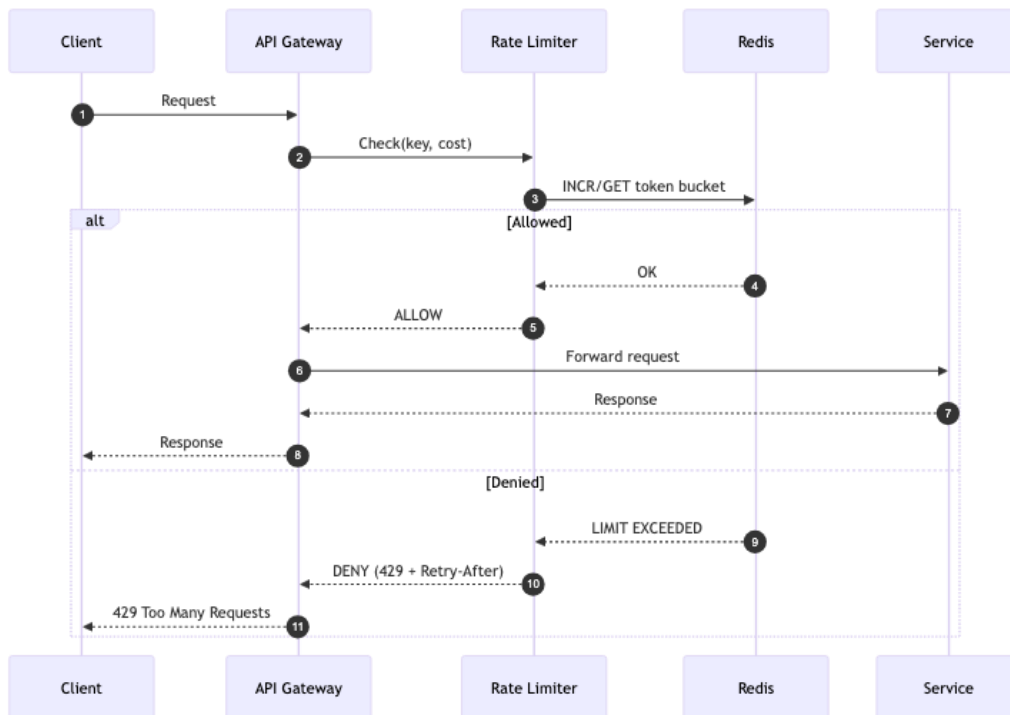
Q02: Design a Rate Limiter (per-user + per-IP)

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Diagrams



Q02 Architecture

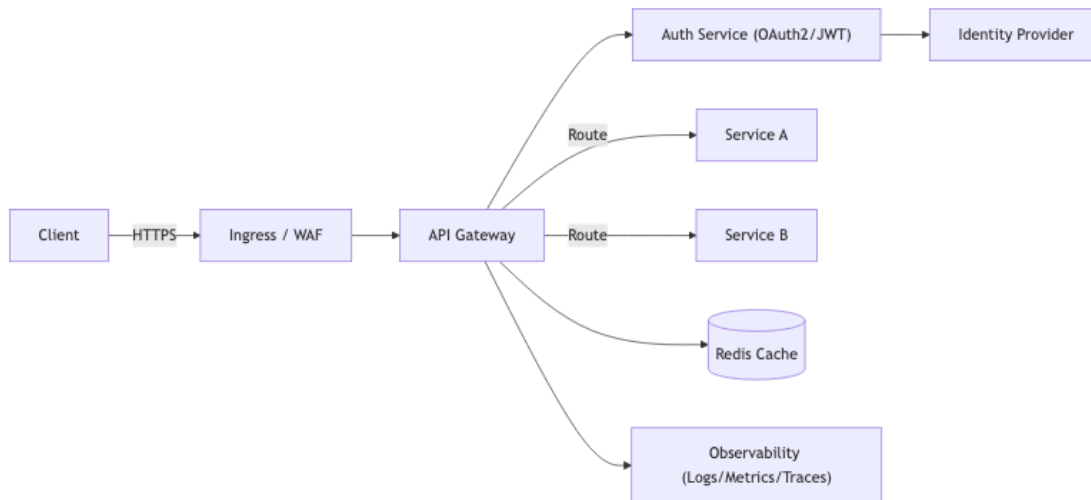


Q02 Key Flow

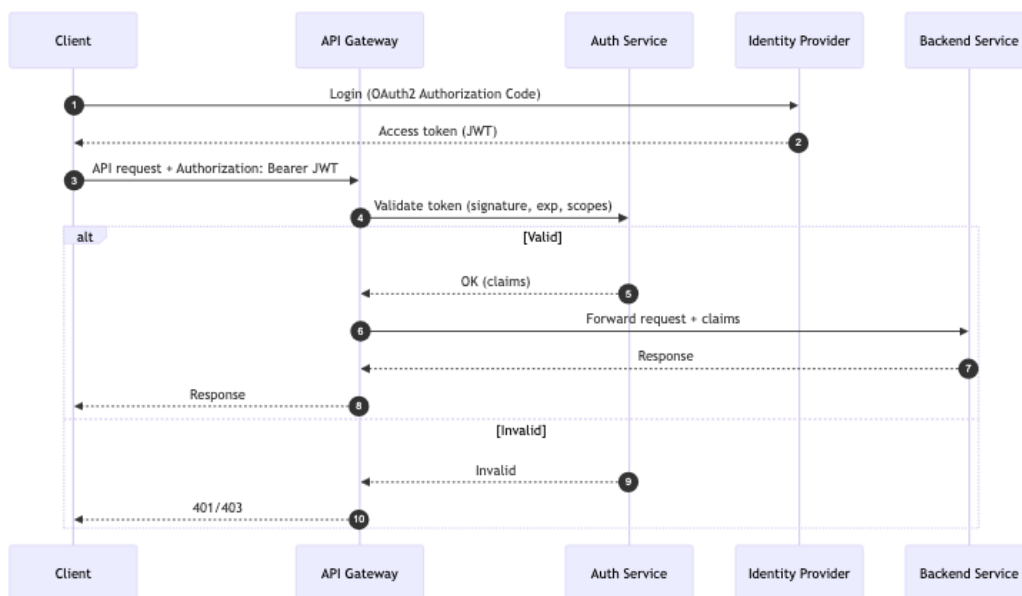
Q03: Design an API Gateway + AuthN/AuthZ (JWT/OAuth2)

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Diagrams



Q03 Architecture

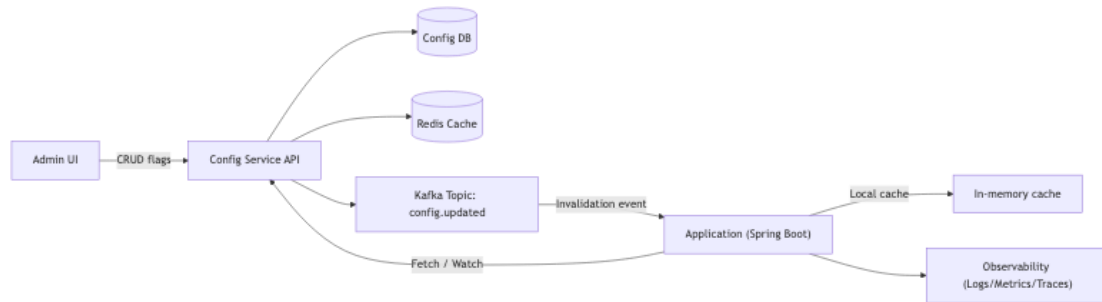


Q03 Key Flow

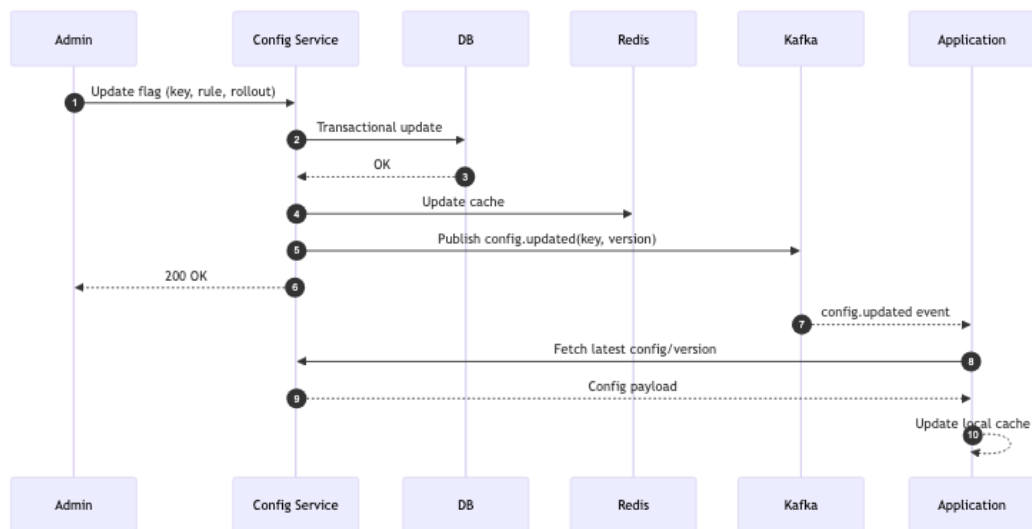
Q04: Design a Feature Flag / Config Service

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Diagrams



Q04 Architecture

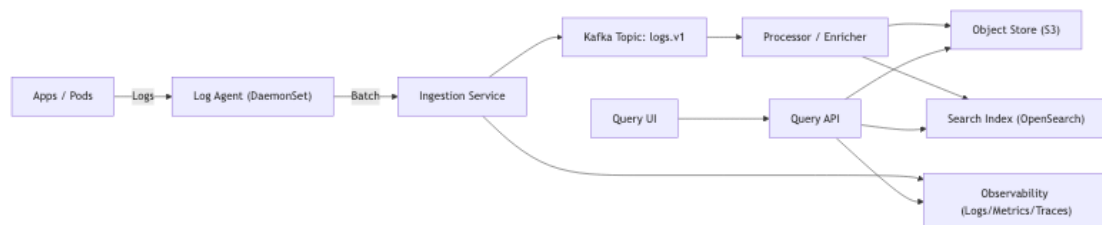


Q04 Key Flow

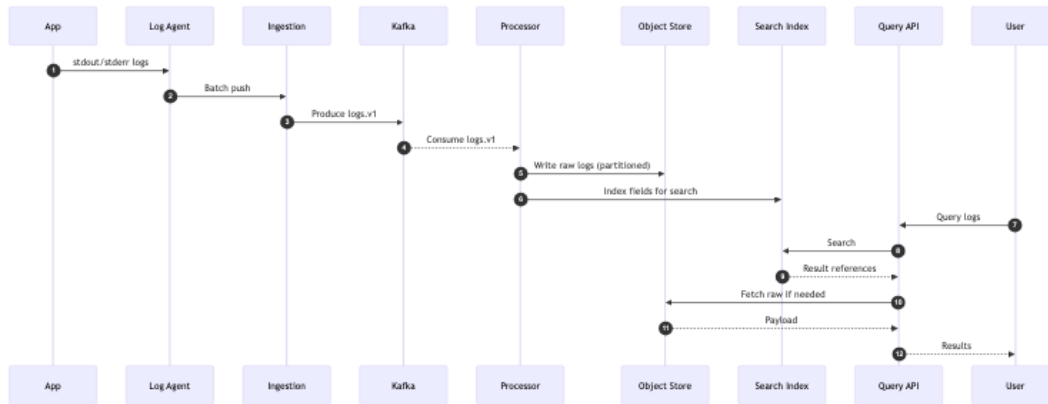
Q05: Design a Logging Platform (collection → storage → search)

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Diagrams



Q05 Architecture



Q05 Key Flow

Q06: Design a Metrics/Monitoring Platform (ingestion + aggregation)

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Diagrams - diagrams/q06_arch.png - diagrams/q06_seq.png

Q07: Design a Notification System (email/SMS/push; retries/DLQ)

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Q08: Design an Event Ingestion Platform (Kafka + schema + consumers)

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Q09: Design an Order Processing Pipeline (Kafka choreography)

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Q10: Design Outbox + CDC Integration Pattern

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Q11: Design Payment Events + Reconciliation Pipeline

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Q12: Design a Fraud Detection Stream (near real-time scoring)

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Q13: Design an E-commerce Checkout + Order System (Saga)

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Q14: Design a Wallet / Ledger System

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Q15: Design a Ticket Booking System (avoid oversell)

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Q17: Design a Subscription Billing System

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Q21: Design a Real-time Collaboration System (high-level)

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Q22: Design Ride Dispatch/Matching (geo + scaling)

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Q23: Design a Real-time Leaderboard

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Q28: Design Location-based Nearby Search (geo index)

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Q29: Design Analytics Clickstream Pipeline

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Diagrams - diagrams/q29_arch.png - diagrams/q29_seq.png

Q30: Design Multi-tenant SaaS Platform Baseline

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Diagrams - diagrams/q30_arch.png - diagrams/q30_seq.png

Cross-cutting Cheat Sheets

Kafka Cheat Sheet

Kubernetes Cheat Sheet

Databases Cheat Sheet

Caching Cheat Sheet

SLO/SLI + Observability Cheat Sheet

Security Cheat Sheet

Diagram Conventions

Naming

- qNN_arch.png for architecture
- qNN_seq.png for key flow/sequence

Recommended Diagram Types

- Architecture: component diagram
 - Flow: sequence diagram
-

Export Commands

Folder Structure

- system-design-master.md
- diagrams/
- assets/
- exports/

Pandoc (DOCX)

- Input: system-design-master.md
- Output: exports/System_Design_Master.docx

Pandoc (PDF)

- Input: system-design-master.md
- Output: exports/System_Design_Master.pdf

Diagram Generation (Mermaid → Images)

- Keep Mermaid source files under diagrams-src/ (optional)
- Export images into diagrams/